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Exploring Sentiment Analysis with Custom and Pre-built BERT Models on IMDb Reviews

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1. Abstract

This paper presents the development of a machine learning-based sentiment analysis tool designed for movie reviews. Our objective was to compare two distinct approaches: a traditional machine learning model and a state-of-the-art deep learning model. We implemented two models for this purpose: a logistic regression model for binary classification, which served as our baseline, and a fine-tuned Bidirectional Encoder Representation from Transformers (BERT). Our comparative analysis revealed that the pre-trained and fine-tuned BERT model outperformed the logistic regression model in terms of both accuracy and efficiency. These findings underscore the advantages of leveraging advanced pre-trained models for complex natural language processing tasks. Additionally, this research provides valuable insights that could be applicable to the development of automated agents for commercial environments.

2. Introduction & Motivation

In his landmark book titled "Influence", American psychologist Robert Cialdini puts 'social proof' as one of seven critical factors that drive change in human behavior (Cialdini, 2001). Simply put, individuals look to the opinions and actions of other people to make personal and economic decisions.

Humans are inherently social beings. Long before the advent of the internet, people relied on word-of-mouth recommendations, and used to ask their friends about the best mechanic, or the best dentist. The web and location-enabled mobile technologies have ushered in an era of explosive growth in user-generated content. Today netizens share their opinions and experiences at scale through product, place, and movie reviews, opinions in the form of vlogs and blogs, and social media content a vast array of topics–from DIY repairs and shopping tips to travel guides and restaurant recommendations.

Studies have shown that online reviews have a significant influence on purchase intent and consumer behavior (Qiu & Zhong, 2024). As consumers become more aware of and reliant on reviews and recommendations, it poses a challenge to businesses–how to manage this deluge of information, especially the kind that may provoke a negative backlash, at the same scale as it is being produced. For example, if a fast food brand runs a national chain of restaurants, how does one keep up with all of the reviews that users are leaving on a daily basis.

Sentiment analysis has implications for multiple domains. The sentiment expressed in a product review on an e-commerce website can influence the revenue of the company. Social media responses to a political announcement wield substantial influence over shaping public policy. In the financial markets, terms such as 'bullish' and 'bearish' are directly related to market sentiment. Bullish indicates a positive sentiment or optimism about the market or a specific stock. Bearish, on the other hand, reflects a negative sentiment or pessimism about the market or a stock.

Hence, sentiment analysis becomes a critical tool for businesses intelligence (Amazon Web Services, n.d.), politicians, and investors to measure collective sentiment, enabling them to make informed decisions and align their strategies with the stakeholder perception.

3. Research Background and Literature Review

Sentiment analysis is a hard problem. Consider the following line from a fictitious movie review:

The soundtrack was insane!

When read literally, the above statement might sound like a negative comment. However, fluent speakers of the English language will understand that "insane" in slang often means "amazing" or "outstanding" depending on context and tone. Sentiment analysis is a challenging problem due to the complexity of human language. Languages are not merely collections of individually defined words; the meaning of words often depends on how they are used and the context in which they are spoken, which can shift their sentiment from one type to another.

Similarly, consider another example from an actual review:

This movie redefines suspense. I kept wondering why I bought the ticket.

The above examples demonstrate that measuring sentiment is simply not a task of assigning positive or negative numbers to respective terms and then counting them at the end. This paper wants to explore different approaches to solve the sentiment analysis problem.

This section captures a brief history of sentiment analysis research approaches. The roots of sentiment analysis can be traced back to the early days of computational linguistics, psychology, and information theory studies, well before computer scientists started to formalize methods and models in later decades. In the early 1950's, Claude Shannon introduced and utilized the concept of grams as in diagram, trigram, and quadragram to address the problem of redundancy of words in the english language and devise compression techniques for electronic signals (1951). In 1957, Osgood et al. conducted a study to quantitatively measure the subjective meanings of words and identified emotional dimensions underlying the meanings of concepts (1957). In 1967, Brown Corpus was published and it stood out as the first comprehensively structured database setting the bar for linguistics research using frequency and distribution of words ("Brown Corpus", 2024).

At a high level, the past work in sentiment analysis can be divided into three phases: heuristic, knowledge, and rules-based, machine learning-based, and deep learning-based methods.

3.1. Heuristic, Knowledge, and Rules-based Approaches

Without taking a machine learning-based approach, sentiment analysis can be tackled using a number of techniques including, but not limited to, heuristic-based, knowledge-based, rules-based, and lexicon-based methods. These approaches have relied heavily on human-designed rules and resources to infer sentiment. Earlier lexicon-based methods relied on predefined dictionaries of sentiment-bearing words, coupled with heuristic shortcuts such as counting positive and negative words to assign scores to text (Stone et al., 1966; Caren, 2019). Knowledge-based approaches leveraged structured resources like WordNet or other corpuses to infer sentiment from semantic relationships (Miller et al., 1990; Hatzivassiloglou & Mckeown, 1997). Rules-based systems used explicit logical conditions, such as character point-of-view (POV) markers that

identify subjectivity to classify sentiment (Wiebe, 1994). While these methods were interpretable and effective for smaller datasets, they often struggled with nuanced language, context, and scalability, paving the way for more data-driven machine learning approaches.

3.2. Early Machine Learning-based Approaches:

The rise of early machine learning introduced a paradigm shift in sentiment analysis, moving from manually crafted rules to data-driven models. Decision Trees were used to detect flames (inflammatory comments) online (Spertus, 1997). Using Naive Bayes, sentiment analysis was correlated with Yelp restaurant ratings reaching more than 75% accuracy (Joyce & Deng, 2019). Support Vector Machines (SVM) were used to bring together disparate sources of information to yield better performance (Mullen & Collier, 2004). Social media sentiment was analyzed using both SVM and Logistic Regression techniques where logistic regression outperformed in both accuracy and precision (Prabhat & Khullar, 2017). These models were often combined with feature engineering techniques, including n-grams, part-of-speech tagging, and syntactic parsing, to improve classification accuracy. Unlike non-ML approaches, these models could generalize better from data, reducing the reliance on handcrafted rules, and outperform human-produced methodologies (Pang et al., 2002). However, their performance was limited by their dependence on carefully designed features and an inability to capture complex relationships in text, which deep learning would later address.

3.3. Deep Learning-based Approaches:

Deep learning has revolutionized sentiment analysis by enabling models to learn hierarchical and contextual representations of text directly from raw data. Convolutional Neural Networks (CNNs) had a lot of success with image processing and computer vision because of their powerful ability to extract features. Initially, CNN based sentiment analysis was shown to be more accurate than the traditional machine learning techniques (Liao et al., 2017), but CNN based methods had their limitations. CNNs lack sequential memory and they cannot capture long-term dependencies or the contextual relationships between words that occur earlier or later in the text. Recurrent Neural Networks (RNNs), alongside Long-Short Term Memory (LSTM) networks, as well as Gated Recurrent Unit (GRU) have shown remarkable ability to model sequential text for sentiment classification (Mienye et al., 2024). Perhaps the most exciting advancement in deep learning literature has been the introduction of attention mechanisms and transformer models (Vaswani et al., 2017), particularly Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019), that has further advanced the field by capturing bidirectional and contextual information. These models eliminate the need for extensive feature engineering, as they automatically learn relevant features from large datasets. Deep learning-based approaches have achieved state-of-the-art performance on a wide range of sentiment analysis tasks, handling complex linguistic phenomena such as sarcasm (Eke et al., 2021), idioms (Abarna et al., 2022), and context-dependent meanings with unprecedented accuracy. Despite their success, challenges remain, including high computational costs (Thompson et al., 2020) and interpretability concerns.

4. Data and Sources

We are using the Large Movie Review Dataset first published by Maas et al. (2011) at Stanford University. This dataset is meant for binary sentiment analysis and comprises substantially more data than the

previous benchmark datasets. This dataset comprises movie reviews with their associated binary classification sentiment labels. The dataset contains 50,000 reviews divided into two even sets of 25,000 each for training and testing. Negative and positive reviews are equal in number. Having an equal number of positive and negative reviews ensures balanced training and prevents biases towards either sentiment. It also facilitates model to model comparison by supporting a more accurate evaluation. It also lowers biases and promotes a fair analysis.

To avoid correlation ratings, there are no more than 30 reviews for any given movie. At the same time, the movies in the positive reviews group are different from the movies in the negative reviews group. Negative reviews were chosen with scores of <=4 out of 10, positive reviews had a score of >=7 out of 10, and the neutral reviews were not included in the dataset. The dataset consists of two directories:

- train/
    - pos/
        - [[id]_[rating].txt]
    - neg/
        - [[id]_[rating].txt]
- test/
    - pos/
        - [[id]_[rating].txt]
    - neg/
        - [[id]_[rating].txt]

Each text file follows the naming convention [[id]_[rating].txt], where id is a unique id and rating is the start rating for that review on a 1-10 scale. For example, a file with a path train/pos/100_8.txt contains a positive labeled train set example with unique id 100 and start rating 8/10 from IMDb.

To streamline data usage, we took the above dataset and processed the files to combine positive and negative reviews into a single column, and added a second column with binary labels 0 and 1 for negative and positive ratings.

```
df.head(10)
```

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive
5	Probably my all-time favorite movie, a story o...	positive
6	I sure would like to see a resurrection of a u...	positive
7	This show was an amazing, fresh & innovative i...	negative
8	Encouraged by the positive comments about this...	negative
9	If you like original gut wrenching laughter yo...	positive

Figure 1. Word Cloud of the Unprocessed Positive Reviews

We have also combined the test and train reviews and will split them using the train_test_split() utility in scikit-learn.

5. Exploratory Data Analysis

Exploratory data analysis is about looking at data, with an open mind, to see what it seems to say (Tukey, 1977). Before we begin any work with our data, including clean-up, analysis, and algorithm training, we can use several techniques to get a general understanding of what we have.

We began by checking the shape and size of our dataset. We used the following commands:

print(df.shape)
print(df[‘sentiment’].value_counts())

We got the following output:

```
(50000, 2)
sentiment
positive    25000
negative    25000
Name: count, dtype: int64
```

Next, we moved to visualize the distribution of sentiments and check the balance between positive and negative reviews.

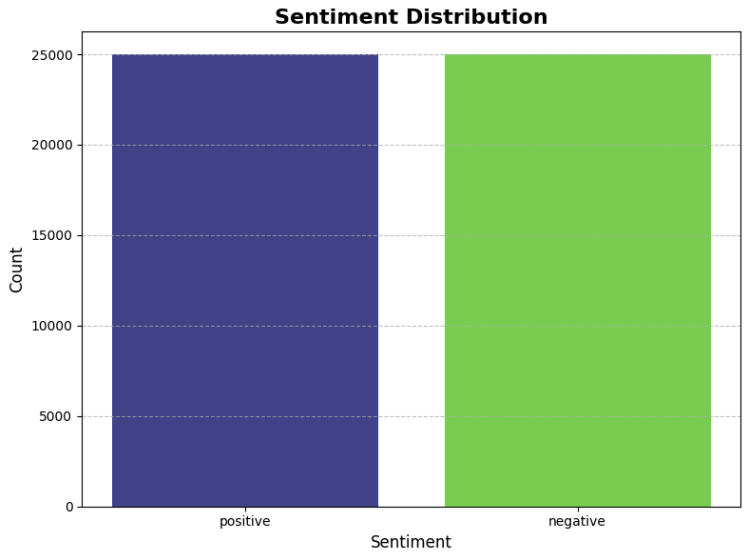


Figure 2: Distribution of Sentiment

As we can verify in **Figure 2** that our data is balanced between positive and negative reviews. This will help us to compare model performances in a better way. Next, we looked at the length of the reviews:

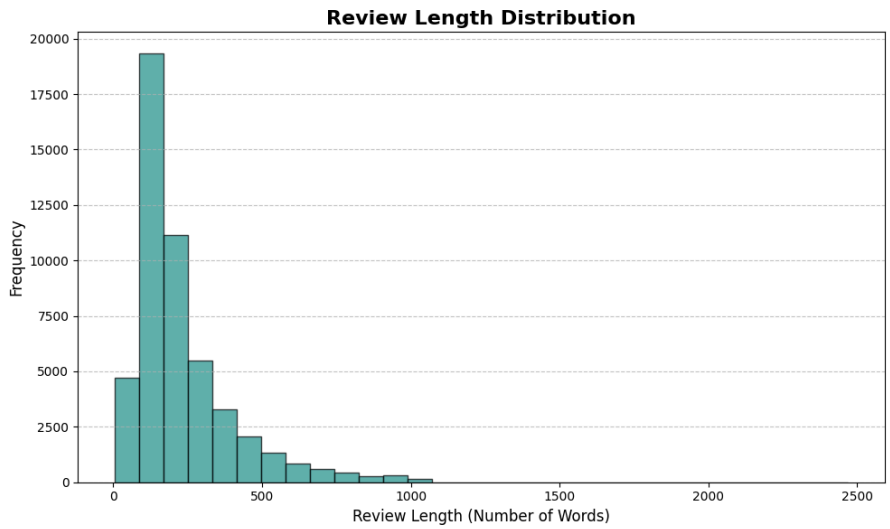


Figure 3: Review Length Distribution

It seems from **Figure 3** that the bulk of our reviews are around 200-300 words. We do have very few longer reviews that are up to 1000 words or slightly more. Next, we look at the top bigrams in positive and negative reviews separately.

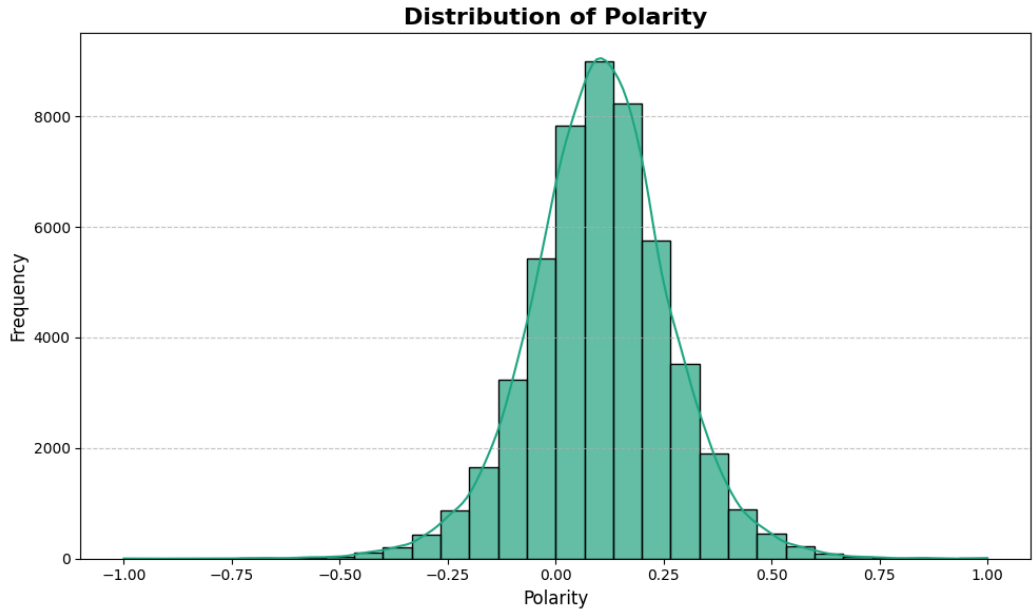


Figure 4: Review Length Distribution

Our distribution of polarity is very slightly skewed towards the positive side (**Figure 4**), which may mean that there are more positive reviews. It could also mean that more of the neutral reviews have been labeled as positive.

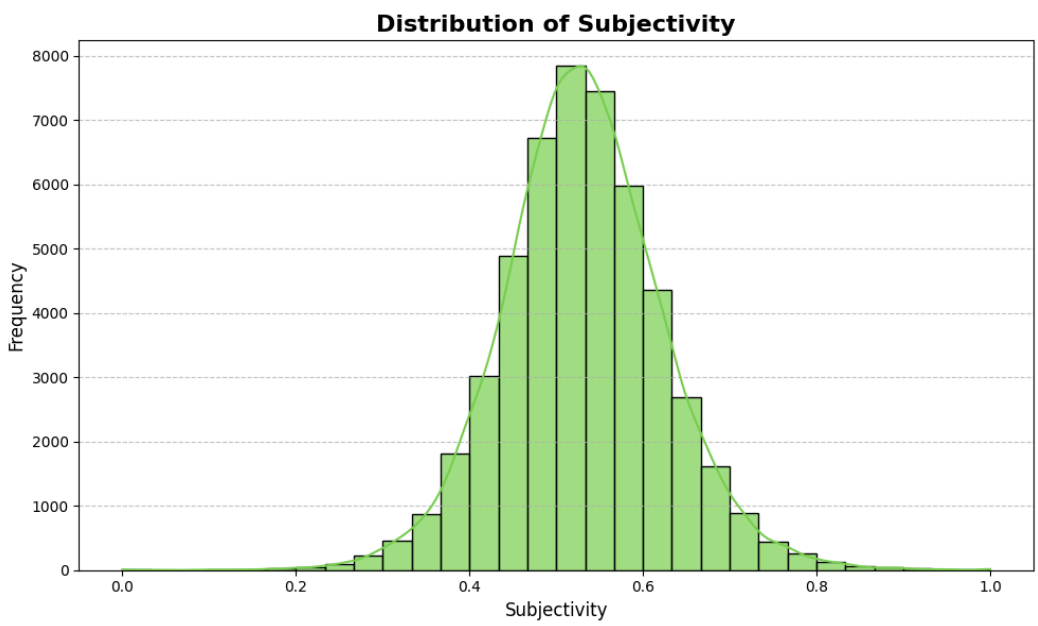


Figure 5: Distribution of Subjectivity

The subjectivity of our data is skewed even a bit more towards the positive side (**Figure 5**). This tells us that reviews have a mix of objective and subjective opinions.

Valence Aware Dictionary and sEntiment Reasoner (VADER) is a lexicon and rule-based sentiment analysis software designed for social media sentiment analysis (Hutto & Gilbert, 2015). Just before we start processing our data, we would like to get a 'second opinion' from VADER.

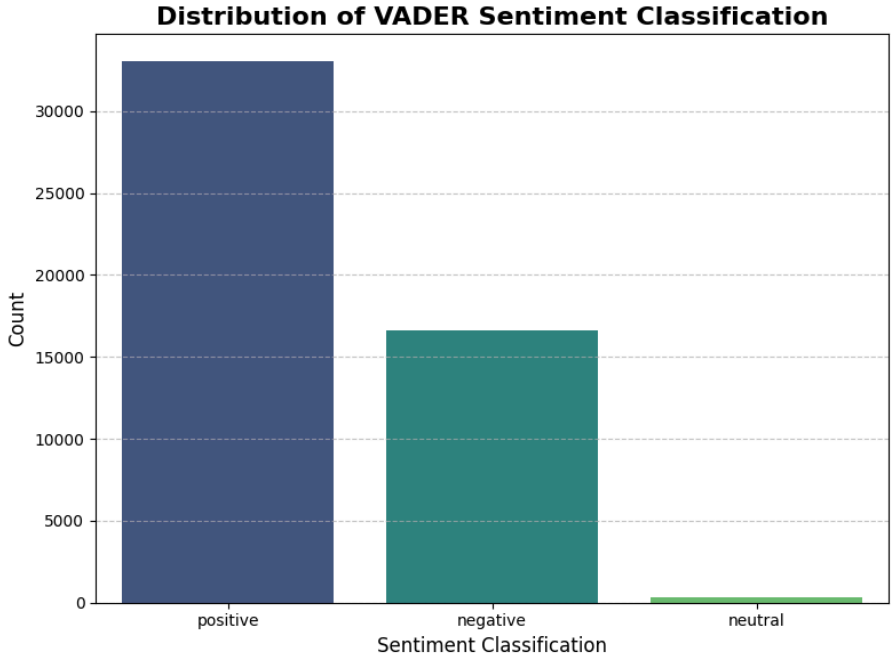


Figure 6: Distribution of VADER Sentiment Classification

We know from our past EDA that we are heavier on the positive reviews and the fact that VADER is counting twice as many positive reviews than negative, could be because of a number of reasons (**Figure 6 & 7**). We believe that VADER is doing so because it is not able to properly classify negative sentiment when it is said in a sarcastic tone while using positive terms (Boghe, 2020)

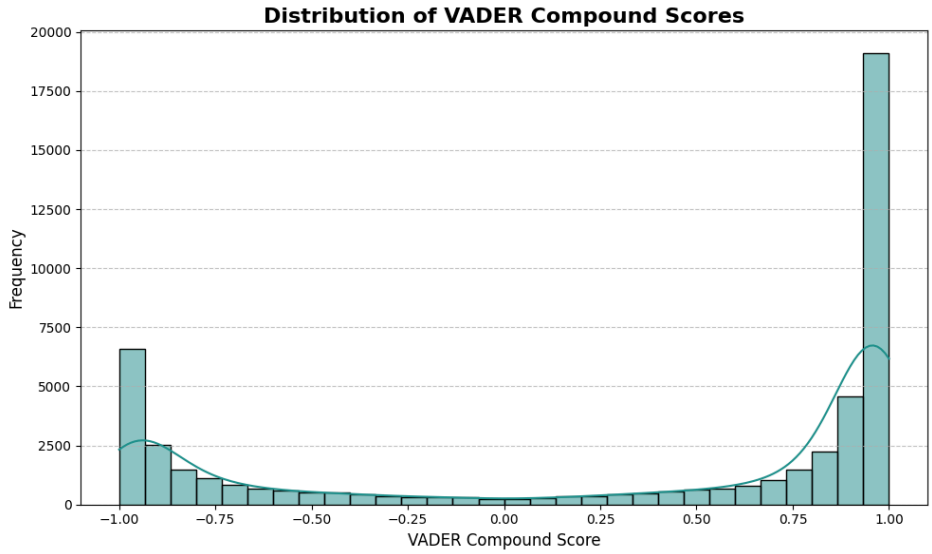


Figure 7: Distribution of VADER Compound Scores

6. Data Pre-Processing

We now come to our next step: Data Pre-processing. Getting the data ready to be consumed by machine learning algorithms is a vital step in natural language processing. This preparation may include the a number of following processes:

1. Removing HTML tags and noisy text
2. Removing special characters
3. Tokenization
4. Removing stop words
5. Lemmatization

We must also mention that the pre-processed data was used only for the Logistic Regression model. Our other model, BERT, did not require as much data pre-processing.

Remove HTML Tags:

As a first step, we proceeded to remove the html tags from our dataset to make our data more meaningful. At the same time, we will remove other unwanted elements like hashtags etc. which don't serve us. For example, the second entry in our dataset is processed as follows:

Unprocessed Text: A wonderful little production.

The filming technique is very unassuming- very old-time-B…

Processed Text: A wonderful little production. The filming technique is very unassuming- very old-time-B…

Remove Special Characters:

Special characters such as punctuation marks, symbols etc. don't aid us in understanding the sentiment. Any superfluous character, for that matter, ends up taking more compute resources and adds to the execution time. Moreover, we would like the algorithm to treat upper and lower characters in the same manner. 'Wonderful' should be treated as equal to 'wonderful'. For this reason we also convert all characters into lowercase. Here is what our pre- and post-processed data might look like at this stage:

Unprocessed Text: A wonderful little production. The filming technique is very unassuming- very old-time-BBC…

Processed Text: a wonderful little production the filming technique is very unassuming very oldtimebbc…

Tokenization:

Studies have shown that accurate tokenization allows for better part-of-speech (POS) tagging, and preserving multiword expressions improves machine translation (Chai, 2022).Tokenization is the step where we extract every word into its own unique unit ready for vectorization. There are no set rules for tokenization and it is highly context dependent. For example, consider the word phrase 'on-the-fly'. A tokenizer may decide to use

the hyphens as delimiter and tokenize the phrase as ‘on’, ‘the’, ‘fly ', but if the context has to be maintained, the tokenization might be done differently.

Remove Stop Words:

Stop words are commonly referred to as prepositions and articles in parts of speech. These include frequently occurring words like ‘the’, ‘of’, ‘at’, ‘in’, ‘and’, etc. Since they occur so frequently, they do add to the computing load. Yet at the same time, they don’t always provide meaningful information to the learning algorithm. They don’t carry significant semantic value, and offer limited value in language processing. They are called stop words because they are ‘stopped’ or filtered out from further processing.

Here is what our pre- and post-processed would look like:

Un-Processed Text: a wonderful little production the filming technique is very unassuming very oldtimebbc…

Processed Text: wonderful little production filming technique very unassuming very oldtimebbc…

Stemming and Lemmatization:

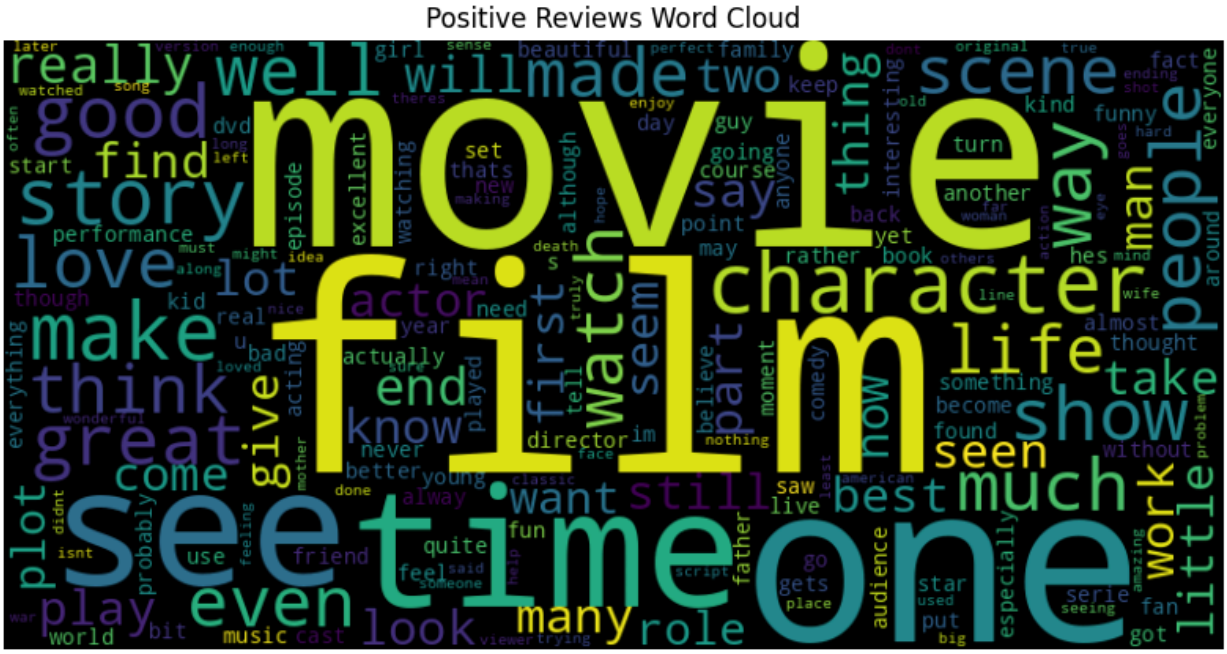
Stemming refers to reducing a word down to its core essence. The reduced form after stemming may or may not be an actual word. For example, ‘Universal’ could become ‘Univers’. Lemmatization, on the other hand, refers to reducing a word to its most basic root form which is still a proper word in the natural language. We chose to lemmatize our data due to the fact that lemmatization produces higher accuracy (Khyani et al., 2021).

Here is the pre- and post-processed comparison:

Un-Processed Text: wonderful little production filming technique very unassuming very oldtimebbc…

Processed Text: wonderful little production film technique very unassuming very oldtimebbc…

To test our pre-processing, we create word clouds as well as top bigram lists in **Figure 8** through **Figure 11**. We can observe that our data does not include any html tags, punctuations, and stop words.


Positive Reviews Word Cloud
movie
film
see
time
one
character
people
story
good
really
well
will
made
two
scene
way
man
find
love
make
think
great
life
show
seen
much
work
little
take
watch
first
seem
part
now
actor
end
know
give
come
plot
play
even
look
many
role
music
world
best
want
still
lot
thing
say
point
turn
another
back
yet
book
rather
actually
never
director
better
quite
friend
use
probably
fun
audience
star
put
especially
serie
fan
got
seeing
place
gets
father
saw
live
nothing
comedy
moment
believe
course
going
anyone
interesting
funny
kind
everyone
around
almost
thought
without
problem
wife
line
hes
mind
action
far
woman
death
truly
tell
face
classic
american
script
viewer
trying
big
used
put
cast
bit
war
feeling
mother
didnt
isnt
amazing
fact
ending
shot
goes
hard
eye
dont
original
true
enough
version
later
watched
song
often
start
dvd
long
left
episode
excellent
watching
performance
must
might
idea
along
though
kid
real
nice
u
bad
loved
acting
everything
wonderful
done
always
young
classic
sure
mean
year
need
right
s
truly
tell
played
director
im
face

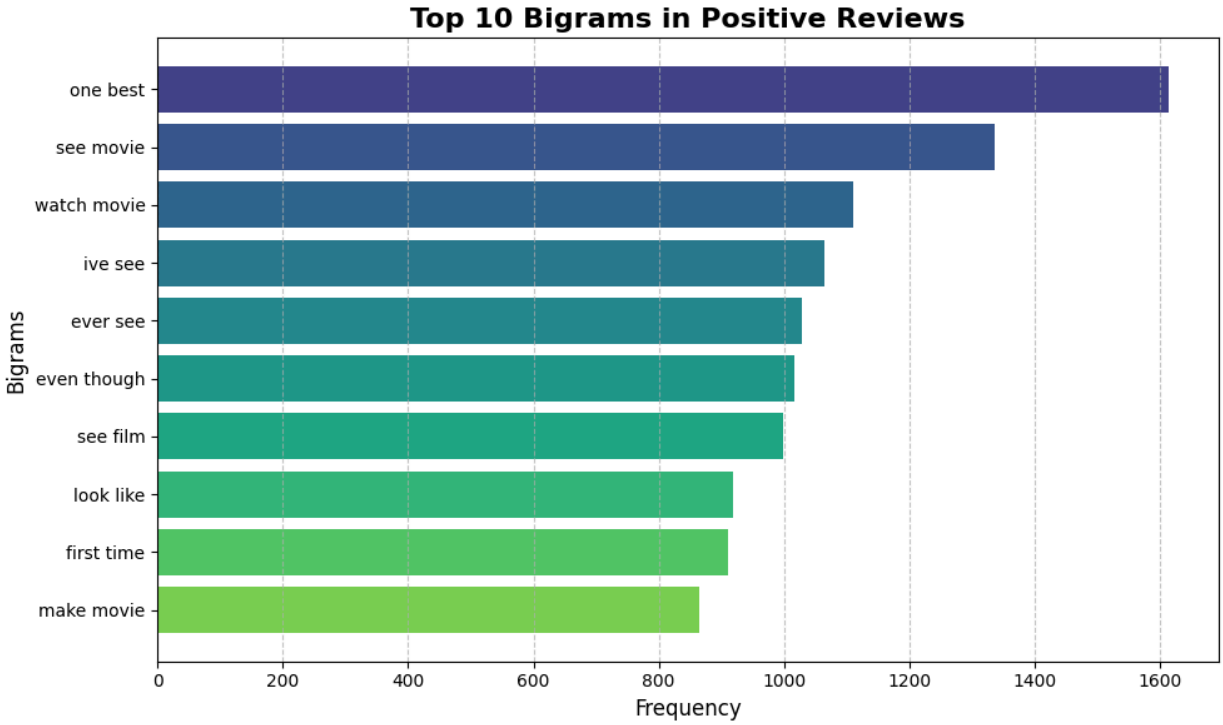


Figure 10: Top 10 Bigrams in Positive Reviews

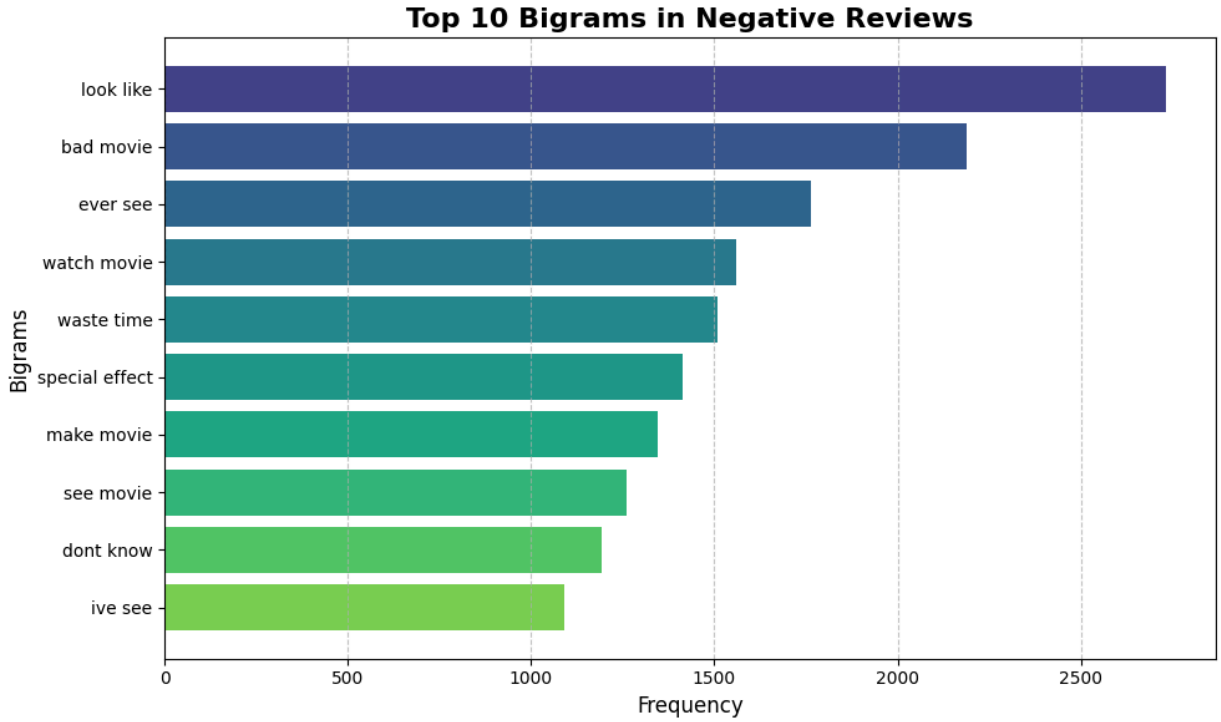


Figure 11: Top 10 Bigrams in Negative Reviews

Vectorization

When it comes to natural language processing, data needs to be transformed into a format that can be processed by machine learning algorithms. This process is usually referred to as vectorization and is a critical

step in an NLP pipeline. There are several techniques available such as Bag of Words (BoW), One-Hot Encoding (OHE), and Term Frequency-Inverse Document Frequency (TF-IDF). We considered all three of them for this project and chose TF-IDF. Here is a brief description of the three:

Bag of Words (BoW):

A text object (e.g. a sentence etc.) can be referred to as a document. The collection of all documents is called the corpus (Tazzyman, n.d.). The set of all words used in the corpus is called the vocabulary. The BoW technique works by converting each document into an n-dimensional vector, where each dimension contains the frequency of every word as it appears in the document.

Pros: BoW is simple and straightforward.

Cons: Usually results in very high-dimensional sparse vectors especially with large vocabularies. Also, BoW does not capture the context of the words.

One-Hot Encoding (OHE):

OHE works by creating binary vectors corresponding to the presence or absence of words int the vocabulary.

Pros: Easy to create.

Cons: Like BoW, OHE also creates high dimensional sparse vectors that are computationally inefficient. All words are treated equally without the context.

Term Frequency-Inverse Document Frequency (TF-IDF):

TF-IDF measures the importance of a word by starting with its frequency in a document, and offsetting it by the frequency of the same word in the corpus. This way, it reduces the weight of terms that occur frequently across documents, but not so frequently in the corpus.

Pros: TF-IDF diminishes the effect of terms that are less important for sentiment analysis, but enhances the effect of the words that might be crucial for a specific comment.

Cons: Still results in sparsity causing high dimensionality.

7. Models

Logistic Regression:

Logistic Regression is a statistical method mostly used for predicting binary outcomes. It's particularly effective in fields such as finance, healthcare, and more pertinently in natural language processing (NLP) for sentiment analysis, where it classifies text data into categories like positive or negative sentiments.

Logistic Regression was selected as a baseline model for several reasons. Its straightforward implementation and the interpretability of its results make Logistic Regression an excellent baseline for benchmarking more complex models. Logistic Regression is relatively quick to train on large datasets, essential for initial exploratory analysis. Lastly, it can provide strong performance in binary classification tasks, particularly when the input features are well-prepared.

BERT:

BERT (Bidirectional Encoder Representations from Transformers) is a state-of-the-art model in natural language processing that has revolutionized the way machines understand human language. Developed by Google, BERT's architecture allows it to understand the context of a word based on all of its surroundings rather than just the words that precede it.

We chose BERT because its ability to process words in context dramatically improves its capacity to understand nuanced language used in movie reviews, which is critical for accurate sentiment analysis. Using a pre-trained model allows us to utilize a network that has already learned a vast amount of information about the language, making it more efficient and effective at a specialized task like sentiment analysis. Unlike traditional models that require extensive feature engineering, BERT can be fine-tuned with just one additional output layer, making it adaptable to the specific nuances of sentiment analysis without extensive preprocessing.

A significant advantage of using BERT is its requirement for minimal preprocessing. Unlike traditional machine learning models where extensive cleaning and normalization of the text are necessary (e.g., removing stopwords, stemming), BERT works effectively with the raw text as it utilizes its own tokenizer to convert text into tokens that it can understand. This tokenizer handles the splitting of words and the addition of special tokens that help the model understand sentence boundaries and relationships between sentences. BERT's tokenizer can manage typical noise in the text, such as special characters and mixed casing, which traditional models would typically require pre-cleaning.

The BERT model used was bert-base-uncased, which processes text in lowercase and has been pre-trained on a large corpus of the English language. The fine-tuning process involved adjusting the BERT model to the sentiment analysis task by adding a classification layer on top of the BERT pre-trained model. The entire model was then trained on the IMDb reviews dataset to adapt its weights to the sentiment classification task specifically.

8. Results

Metric	Logistic Regression	BERT
Accuracy	0.8832	0.9274
Precision (Positive)	0.87	0.91
Precision (Negative)	0.89	0.95
Recall (Positive)	0.90	0.95
Recall (Negative)	0.87	0.90
F1-Score (Positive)	0.89	0.93
F1-Score (Negative)	0.88	0.93

Figure 12: Comparison of Performance Metrics for Logistic Regression and BERT

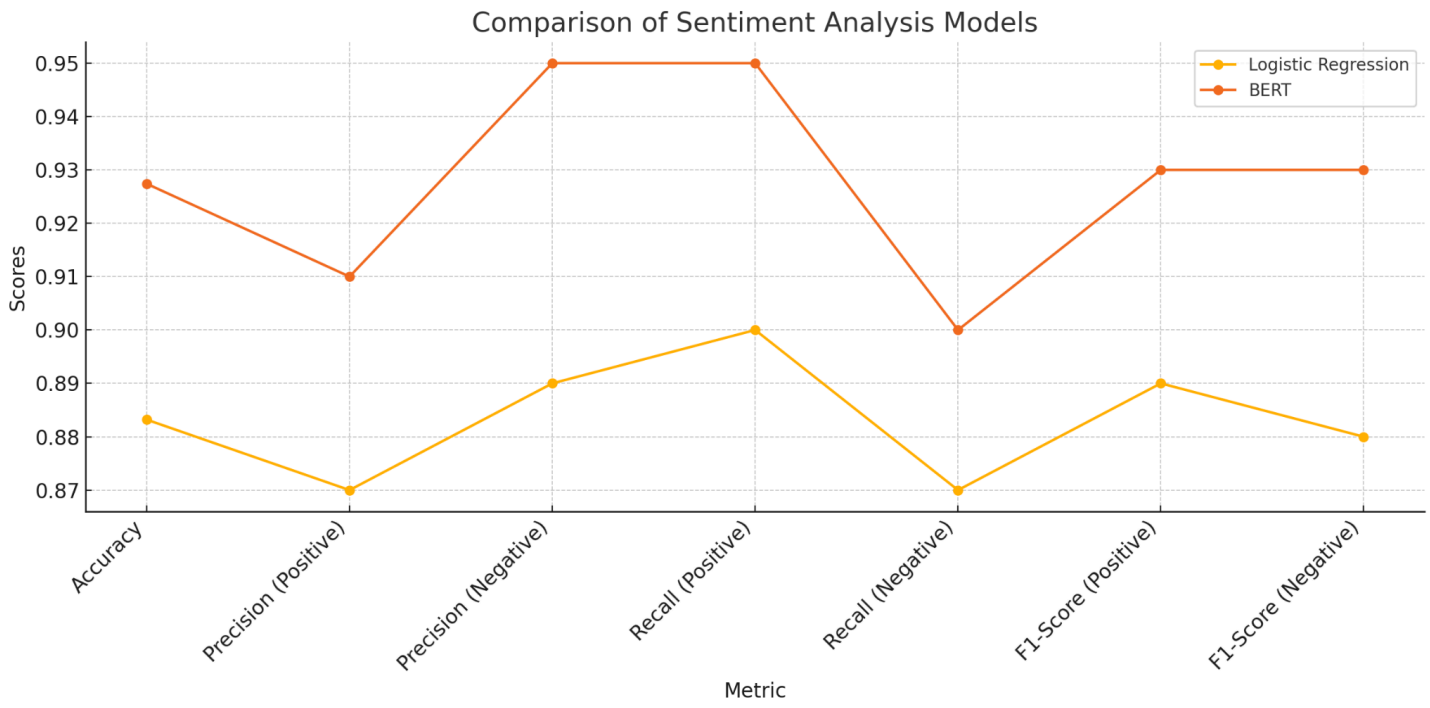


Figure 13: Comparison of Sentiment Analysis Models

9. Discussion

BERT brings a significant advantage over Logistic Regression due to its ability to understand context. By processing text bidirectionally, BERT considers both forward and backward contexts simultaneously. It captures nuanced sentiments better than Logistic Regression, which relies on simpler feature representations. BERT also benefits from pretraining on massive corpora like Wikipedia and BooksCorpus, enabling it to grasp rich linguistic patterns before fine-tuning on specific tasks. Logistic Regression, by contrast, lacks this advantage, depending solely on the provided dataset.

When it comes to handling complex linguistic challenges, BERT's transformer architecture excels at identifying phenomena like negation, sarcasm, and multi-word expressions. Logistic Regression, being a linear model, struggles with these intricacies. BERT also dynamically generates context-sensitive embeddings, offering deeper insights into text, whereas Logistic Regression depends on static features like TF-IDF or bag-of-words. Our metrics further underline BERT's superiority, with higher precision (0.91 positive, 0.95 negative), recall (0.95 positive, 0.90 negative), and F1-scores (0.93 for both), demonstrating its reliability and robustness.

Despite its advantages, BERT comes with trade-offs. It demands greater computational resources, making it less efficient than the lightweight Logistic Regression model. Additionally, implementing BERT is more complex, requiring substantial infrastructure, whereas Logistic Regression is straightforward and easy to deploy. While BERT's deep neural network structure excels at modeling nonlinear relationships, the simplicity of Logistic Regression can still be a practical choice for less demanding tasks.

10. Conclusion

The comparison clearly demonstrates BERT's superior performance in sentiment analysis across all metrics, particularly in precision, recall, and F1-scores. This makes it an ideal choice for tasks where accuracy and robustness are critical, such as customer feedback analysis or social media sentiment tracking.

However, the choice of model should be guided by the use case requirements. For resource-constrained environments or quick prototyping, Logistic Regression remains a viable option. For applications demanding high precision and contextual understanding, BERT is the optimal choice, despite its higher computational costs. This analysis highlights the importance of aligning model selection with the project's objectives and resource constraints.

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